

Lambda-Flight v1.1: Operational and Latent Coherence Transport Models

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Abstract

FRC 787.787 v1.1 separates three objects previously called Lambda. $\text{Lambda_obs} = \text{Lambda} \circ \ln C_obs$ is an operational transform of preregistered observables; Lambda_eq is an observation-derived target; Lambda_dyn is an optional latent reaction-diffusion state that relaxes toward that target. A transport model must select Lambda_obs or separately validated Lambda_dyn before scoring. Support for Lambda_dyn requires improvement over the observation-only model and an equal-feature non-Lambda latent baseline. Even successful residual prediction would not establish a fundamental field. The nested baselines, negative controls, six test cards, and numerical

regime pilot are retained. The pilot calibrates a chosen PDE and does not validate a physical Lambda-Flight mechanism.

1. Scope

FRC 787.787 is a framework paper for coherent transport in driven open physical systems. It is not a claim of a new fundamental force, a replacement for accepted equations, or a license to reinterpret famous experiments after the fact.

The framework asks one testable question: does a measured coherence field improve held-out prediction after the accepted baseline has already been given a fair chance?

If the answer is no, Lambda-Flight fails for that platform. If the answer is yes, the result is a nested residual law, not a discovery that ordinary dynamics were absent.

This is the consistency lock with the 100-series. FRC 100.003 treats microstate-distribution dynamics as one conditional collapse candidate. FRC 100.007 supplies a narrow rho-drift audit, not a replacement ontology. FRC 566.030 computes an exact system-only ledger curve without settling the open physical conjecture. FRC 787.787 therefore keeps every Lambda object operationally typed and baseline-tested.

2. Definitions

Let $\Omega \subset \mathbb{R}^d$ be a physical domain. The minimal measured fields are:

Symbol	Meaning
$\rho(x, t)$	conserved or approximately conserved density
$\psi(x, t)$	order parameter, mode amplitude, phase field, or envelope
$E(t)$	external drive or control parameter
$C_{\text{obs}}(x, t) > 0$	positive dimensionless coherence functional built from preregistered observables
$\Lambda_{\text{obs}}(x, t)$	instantaneous operational transform
$\Lambda_{\text{eq}}(x, t)$	observation-derived calibrated target
$\Lambda_{\text{dyn}}(x, t)$	optional latent or coarse-grained dynamical state

The operational transform is

$$\Lambda_{\text{obs}}(x, t) = \Lambda_0 \ln C_{\text{obs}}(x, t).$$

A useful construction is multiplicative:

$$C_{\text{obs}}(x, t) = \prod_{j=1}^m C_j(x, t)^{w_j}, \quad w_j \geq 0, \quad \sum_j w_j = 1.$$

Then

$$\frac{\Lambda_{\text{obs}}(x, t)}{\Lambda_0} = \sum_{j=1}^m w_j \ln C_j(x, t).$$

The weights and normalization constants must be fixed before testing or fitted only on training data. A field built after seeing the event is a visualization, not evidence.

The target $\Lambda_{\text{eq}} = \widehat{\mathcal{G}}_{\text{eq}}[O, E]$ is fixed by a declared calibration or training rule. Where finite lag is tested, the optional latent state obeys

$$\tau_{\Lambda}(\partial_t + u \cdot \nabla) \Lambda_{\text{dyn}} = \ell_{\Lambda}^2 \nabla^2 \Lambda_{\text{dyn}} - \Lambda_{\text{dyn}} + \Lambda_{\text{eq}} + \xi_{\Lambda}.$$

Λ_{dyn} is a model state, not a directly observed or fundamental field. A fundamental Lambda field is a separate conjecture requiring independent assumptions and evidence.

3. Coherence Factors

The factors C_j must be platform-specific and measurable. Examples include:

Factor	Example construction	Interpretation
Amplitude regularity	$C_A = \exp[-\text{Var}_{\ell}(A)/A_0^2]$	smoothness of envelope amplitude
Phase order	$C_{\phi} = R_{\ell} + \epsilon$, with R_{ℓ} the local circular phase order	local phase locking
Spectral concentration	$C_s = (P_{\text{band}} + \epsilon)/(P_{\text{total}} + \epsilon)$	mode concentration
Defect suppression	$C_d = \exp[-n_d/n_0]$	low local defect density

Correlation persistence	$C_\xi = \exp[-\ell_0/\xi_{\text{corr}}]$	long correlation length
Dissipation regularity	$C_\sigma = \exp[-\sigma_{\text{exc}}/\sigma_0]$	reduced excess entropy-production proxy
Boundary matching	$C_b = \exp[-M_b^2/b_0^2]$, where M_b is boundary mismatch	compatibility with constraints

No row in this table is mandatory. A platform may require different factors. What is mandatory is that the factors are chosen before the held-out prediction and that simpler single-factor explanations are tested.

4. Operational Lambda-Flight

The residual transport field must be selected before scoring:

$$\Lambda_{\text{tr}} = \begin{cases} \Lambda_{\text{obs}}, & \text{observation-only residual model,} \\ \Lambda_{\text{dyn}}, & \text{separately validated finite-lag model.} \end{cases}$$

Let $q(x, t)$ be the measured density or intensity of a coherent structure. Define its center and radius:

$$X(t) = \frac{\int_{\Omega} x q(x, t) dx}{\int_{\Omega} q(x, t) dx}, \quad R^2(t) = \frac{\int_{\Omega} \|x - X(t)\|^2 q(x, t) dx}{\int_{\Omega} q(x, t) dx}.$$

Let $K(t)$ be an internal coherence score such as phase order, spectral concentration, or inverse defect density. A structure is a Lambda-Flight candidate over $[0, T]$ only if:

1. $K(t)$ stays above a preselected threshold K_{min} for most of $[0, T]$.
2. $R(t)$ grows more slowly than the matched diffusive or dissipative baseline.
3. The residual velocity after known baseline forces are removed is predicted by the precomputed $\nabla \Lambda_{\text{tr}}$ field on held-out data.

The third condition is the load-bearing one. Persistent motion alone is not Lambda-Flight.

5. Minimal Closure

The minimal continuum closure is a nested extension of ordinary transport:

$$\partial_t \rho + \nabla \cdot (\rho u) = \nabla \cdot (D_\rho \nabla \rho - \chi \rho \nabla \Lambda_{\text{tr}}) + S_\rho.$$

The flux is

$$J_\rho = \rho u - D_\rho \nabla \rho + \chi \rho \nabla \Lambda_{\text{tr}}.$$

This equation must be read conservatively. The term $\chi \rho \nabla \Lambda_{\text{tr}}$ is a residual closure. It is permitted only when:

1. the accepted baseline transport model has been specified;
2. its residuals are measured;
3. $\nabla \Lambda_{\text{tr}}$ predicts those residuals out of sample;
4. simpler equal-access alternatives do not explain the same gain.

If these conditions are not met, χ is set to zero and the model reduces to the baseline.

Support for Λ_{dyn} requires improvement over both the observation-only Λ_{obs} model and an equal-feature non-Lambda latent baseline. Otherwise $\Lambda_{\text{tr}} = \Lambda_{\text{obs}}$ or the residual term is removed.

6. Variational Discipline

A phenomenological Lyapunov-compatible form is

$$F[\rho, \psi, \Lambda_{\text{dyn}}] = \int_{\Omega} \left[f_\rho(\rho) + \frac{\kappa_\psi}{2} |\nabla \psi|^2 + U(\psi) + \frac{\kappa_\Lambda}{2} |\nabla \Lambda_{\text{dyn}}|^2 + \frac{a_\Lambda}{2} (\Lambda_{\text{dyn}} - \Lambda_{\text{eq}})^2 - \eta \Lambda_{\text{c}} \right]$$

The original formal draft treated this as a free-energy proof. The corrected status is weaker and cleaner: it is a Lyapunov-compatible template under stated assumptions. If Λ_{eq} depends on ρ or ψ , the full functional derivatives include additional terms. Those terms must be retained in a platform-specific derivation. Without that derivation, the variational section is a discipline check, not a theorem of monotonic free-energy decay.

7. Linear Stability

For the two-field model

$$\partial_t \rho = D \nabla^2 \rho - \chi \nabla \cdot (\rho \nabla \Lambda_{\text{dyn}}),$$

$$\tau \partial_t \Lambda_{\text{dyn}} = \ell^2 \nabla^2 \Lambda_{\text{dyn}} - \Lambda_{\text{dyn}} + a(\rho - \rho_0),$$

linearizing around $(\rho, \Lambda_{\text{dyn}}) = (\rho_0, 0)$ with perturbations $e^{\sigma t + i\mathbf{k} \cdot \mathbf{x}}$ gives

$$(\sigma + Dk^2)(\tau\sigma + 1 + \ell^2 k^2) - \chi a \rho_0 k^2 = 0.$$

Coherence feedback can overcome diffusive smoothing only when

$$\chi a \rho_0 > D(1 + \ell^2 k^2).$$

This is a candidate instability threshold. It becomes a physical claim only after a , χ , and the coherence channel are estimated without test leakage.

8. Flight Numbers

Let L be the structure size, $\Delta\Lambda_{\text{tr}}$ the selected transport-field contrast, and D the effective diffusivity. The coherence transport number is

$$Pe_\Lambda = \frac{v_\Lambda L}{D} \sim \frac{\chi \Delta\Lambda_{\text{tr}}}{D}.$$

For the finite-lag Λ_{dyn} model only, let V be the observed or baseline-predicted structure speed. The relaxation number is

$$St_\Lambda = \frac{\tau_\Lambda V}{L}.$$

The regime map is:

Regime	Expected behavior	Kill condition
$Pe_\Lambda < 1$	diffusion or baseline transport dominates	claimed flight persists with shuffled or zero Lambda
$Pe_\Lambda > 1$, $St_\Lambda \ll 1$	adiabatic coherence-guided transport possible	no held-out residual prediction
$Pe_\Lambda > 1$, $St_\Lambda \sim 1$	lag, overshoot, hysteresis	hysteresis is already explained by baseline
$St_\Lambda \gg 1$	Lambda decouples from moving structure	no degradation relative to low- St_Λ case

The estimates must be made from training data or direct calibration:

Quantity	Required estimate
χ	residual mobility from training trials only
$\Delta\Lambda_{\text{tr}}$	held-out-blind contrast of the preselected transport field
D	matched diffusion or spreading coefficient
τ_{Λ}	response lag from independent relaxation or sweep data
V	baseline speed or training-set structure speed
L	preselected structure scale

9. Nested Baselines

Lambda-Flight is evaluated as a nested residual model. The model stack is:

Level	Model	Purpose
B0	persistence baseline	recent trajectory continues
B1	accepted platform equation	PDE, TDDFT, phase-field, Kuramoto, reaction-diffusion, or other domain baseline
B2	best single observable	amplitude, phase order, defect density, energy, or one coherence factor
B3	equal-feature non-Lambda latent model	same observables but no Lambda geometry or relaxation law
B4	typed Lambda model	$B1 + \Lambda_{\text{tr}}, \nabla\Lambda_{\text{tr}}$ and, only for the lag model, τ_{Λ}

Support requires B4 to beat B0-B3 on held-out data by a preselected metric with uncertainty estimates. A visible Lambda map is not support. A post-hoc fit is not support. A famous experiment that can already be explained by B1 is not support.

10. Negative Controls

Every serious test should include at least these controls:

1. Shuffled Lambda labels across trials.
2. Time-reversed Lambda fields.
3. Spatially shifted Lambda fields.

4. Single-factor ablations.
5. Equal-feature latent baselines.
6. Training/test split by run, parameter regime, or time block.
7. No-refit prediction horizons fixed before scoring.

If the Lambda model survives only when these controls are omitted, it is not a pillar result.

11. Six Test Cards

The six predictions are now written as test cards. Each has a null and a kill clause.

Prediction	Test	Null	Kill
P1 coherence-gradient drift	residual $\dot{X}_{\text{res}}$ aligns with $\nabla \Lambda_{\text{tr}}$	residuals are baseline noise	no held-out alignment after controls
P2 basin residence	T_{res} scales with $\Delta \Lambda_{\text{tr}}$ in a rare-event regime	residence explained by energy or amplitude	single observable beats Lambda
P3 transition early warning	Lambda curvature, variance, or gap degrades before collapse	baseline observable warns equally early	no lead time on held-out transitions
P4 finite-relaxation hysteresis	lag grows with $\tau_{\Lambda} r$ under sweeps	standard response delay explains loop	no Lambda-specific scaling
P5 coherence-over-amplitude selection	higher Lambda structure persists when amplitude is matched	amplitude or energy selects structure	amplitude-only model wins
P6 Pe/St collapse	flight fails outside registered Pe/St window	failure is numerical or baseline instability	Pe/St does not stratify outcomes

12. Anti-Precursor-Hunting Rule

FRC 787.787 must not be used to claim that a new external result confirms FRC because it contains a delay, a coherent packet, a transition, or a transport anomaly. That is precursor hunting.

The rule is:

1. The Lambda observables must be defined before scoring the external result.
2. The accepted baseline must be implemented or quoted as the primary null.

3. The Lambda model must make a separable prediction that the baseline does not already determine.
4. The prediction must be scored out of sample.

The attosecond-STM gate is the example. A measured tunneling delay is not automatically τ_{Λ} . The disciplined question is whether field-to-response delay tracks a pre-registered surface coherence variable better than the TDDFT response baseline. If TDDFT already contains the dependence, there is no FRC paper. If a separable Lambda relaxation law beats TDDFT out of sample, the result can become a separate confirmation-of-existing-prediction paper.

13. Worked Platform Protocol

A controlled first platform should be easier than attosecond STM. Suitable candidates are coupled oscillator lattices, driven electrical resonator arrays, nonlinear optical cavities, reaction-diffusion media, and phase-field simulations with known ground truth.

A minimal protocol:

1. Choose the accepted platform equation and fit it on training data.
2. Define C_j factors from measured present/past observables only.
3. Fix normalization and weights on training data.
4. Compute Λ_{obs} and, if registered, Λ_{dyn} ; select Λ_{tr} ; then compute its gradients and flight numbers.
5. Predict held-out trajectory, transition time, residence time, or collapse class.
6. Score B0-B4 and all negative controls.
7. Publish the failure if B4 does not beat the baselines.

14. Numerical Pilot

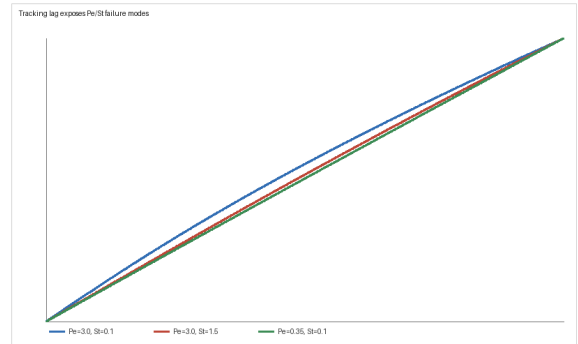
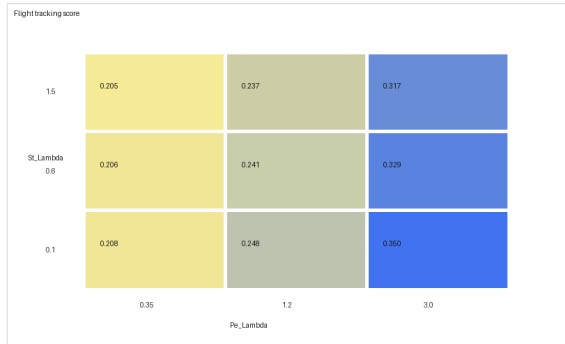
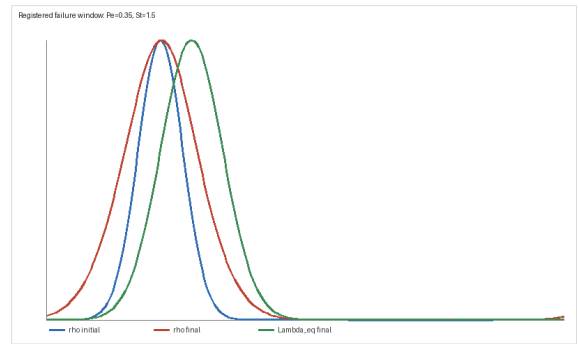
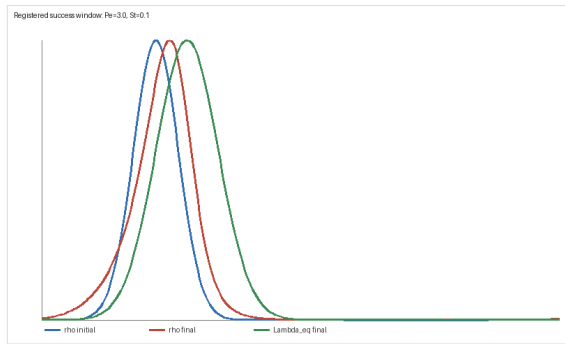
The release package includes a deterministic toy pilot:

```
pilot/pilot_lambda_flight.py
```

The pilot simulates a conserved one-dimensional packet trying to follow a moving coherence basin. It uses a conservative upwind finite-volume flux for density transport and a semi-implicit Fourier update for Lambda relaxation. The sweep varies $Pe_{\Lambda} \in \{0.35, 1.20, 3.00\}$ and $St_{\Lambda} \in \{0.10, 0.60, 1.50\}$.

The code labels $Pe_{\Lambda} \geq 1$ and $St_{\Lambda} \leq 0.6$ as its internal regime window. Because that label is built into the toy model, the pilot is only a solver and regime sanity check; it does not independently validate Prediction 6 or a physical flight threshold.

Metric	Best registered case	Worst case
Pe_{Λ}	3.00	0.35
St_{Λ}	0.10	1.50
median lag / L	0.448	0.499
final basin mass	0.594	0.450
final width ratio	1.388	1.709
tracking score	0.350	0.205



Lambda-flight toy pilot. A conserved packet follows a moving coherence basin best in the high- Pe_{Λ} , low- St_{Λ} corner. The pilot is a numerical regime check, not empirical confirmation.

The pilot gives the right limited lesson: higher Pe_{Λ} improves capture, larger St_{Λ} weakens tracking, and low Pe_{Λ} cannot be rescued by plotting Lambda after the fact. The separation is modest, so the paper should not oversell it.

15. Publication Standard

FRC 787.787 becomes a pillar paper only if its public version contains:

1. the measured-Lambda consistency lock;
2. a strict nested baseline hierarchy;
3. anti-precursor-hunting language;
4. pre-registration cards for each prediction;
5. negative controls;
6. a reproducible pilot or worked platform protocol;
7. clear kill clauses.

This v1.1 package contains those elements at the framework level. It should still be treated as a working preprint until at least one platform test is run against a real accepted baseline.

16. Conclusion

Lambda-Flight is a narrow claim about residual coherent transport. The operational transform

$$\Lambda_{\text{obs}} = \Lambda_0 \ln C_{\text{obs}}$$

turns measured coherence factors into a geometry that can be tested against motion, stability, residence, hysteresis, and collapse. An optional Λ_{dyn} adds finite lag only if it beats both the observation-only model and an equal-feature latent baseline. The framework is useful only when the selected transport field improves held-out prediction beyond accepted baselines and simpler alternatives.

That restriction is the point. If Lambda adds nothing, the platform kills the claim cheaply. If Lambda survives the nested tests, 787.787 becomes a disciplined bridge between the 100-series coherence program and concrete non-equilibrium physical systems.

Even a successful residual model would not establish a fundamental Lambda field.

17. Version note

Version 1.1 separates Λ_{obs} , Λ_{eq} , and Λ_{dyn} ; requires preselection of Λ_{tr} ; and labels the numerical pilot as a solver/regime sanity check rather than threshold evidence.

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